Aim: To perform Graphical Exploratory Data Analysis.

Dataset:

Game Title	Release	Developer	Publisher	Genre	Platform(s)	metacritic_score	estimated_sales	ESRB Rating	Avg. Playtime
Red Dead Redemption 2	2,018	Rockstar Studios	Rockstar Games	Action-Adventure, Open World	PS4, Xbox One, PC,	97	57	M	50
Grand Theft Auto V	2,013	Rockstar North	Rockstar Games	Action-Adventure, Open World	PS4, Xbox One, PC,	96	195	M	32
Resident Evil 7: Biohazard	2,017	Capcom	Capcom	Survival Horror	PS4, Xbox One, PC,	86	13	M	10
Resident Evil Village	2,021	Capcom	Capcom	Survival Horror, Action	PS4, PS5, Xbox, PC	84	10.50	M	11
Call of Duty: Black Ops III	2,015	Treyarch	Activision	First-Person Shooter	PS4, Xbox One, PC	81	26	M	7
Call of Duty: Modern	2,019	Infinity Ward	Activision	First-Person Shooter	PS4, Xbox One, PC	80	30	M	6
Half-Life	1,998	Valve	Valve	FPS, Sci-Fi	PC, PS2, Linux, macOS	96	9.30	M	13
Left 4 Dead	2,008	Valve	Valve	FPS, Survival Horror	PC, Xbox 360	89	12	M	8
Horizon Forbidden West	2,022	Guerrilla Games	Sony Interactive	Action RPG, Open World	PS4, PS5, PC	88	8.40	T	30
Horizon Zero Dawn	2,017	Guerrilla Games	Sony Interactive	Action RPG, Open World	PS4, PC	89	24.30	T	23

Loading the Dataset:

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns

df = pd.read_excel('AIDS-1EXP3.xlsx')

print("Dataset Info:")
print(df.info())
print("\nFirst few rows:")
print(df.head())
print("\nColumn names:")
print(df.columns.tolist())
```

Output:

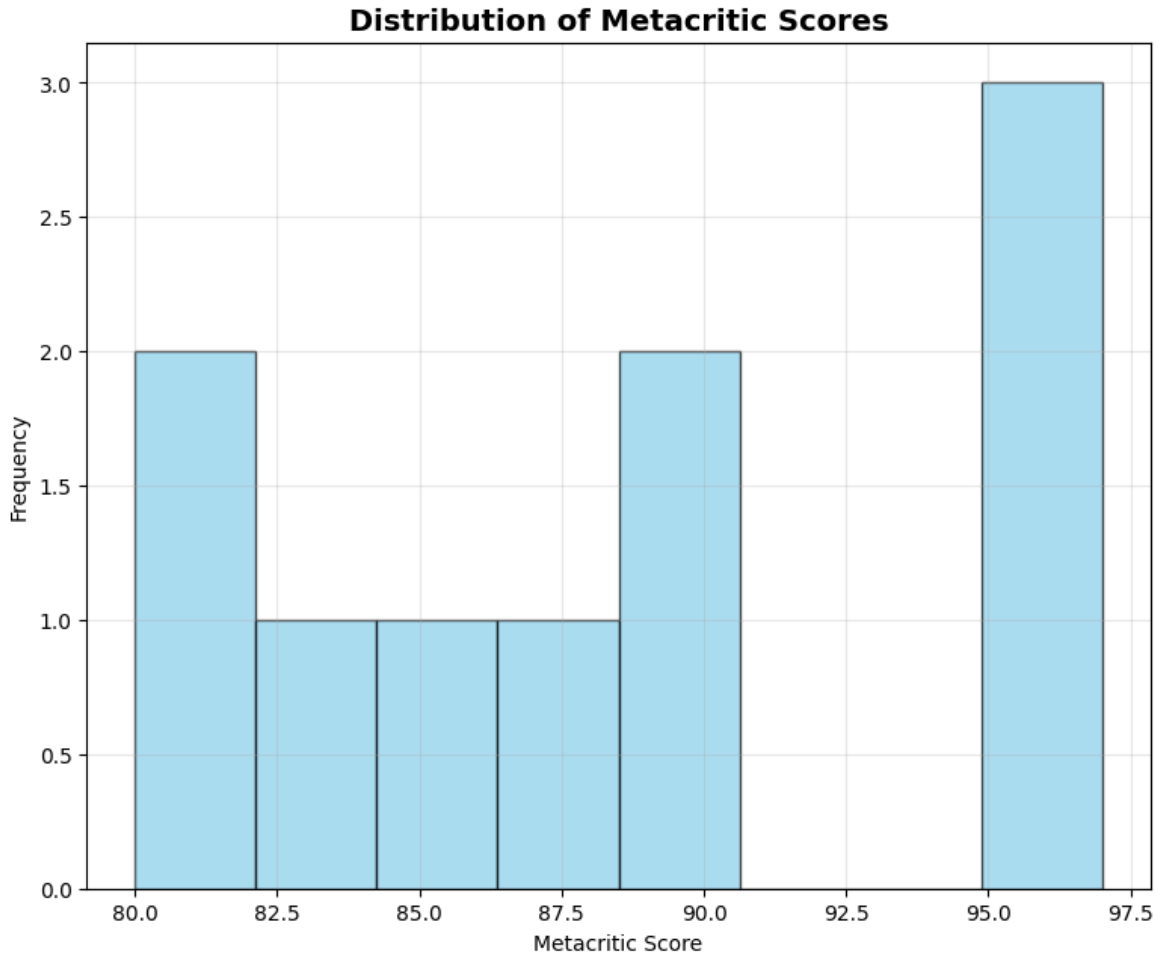
```
Dataset Info:
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 10 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Game Title                    10 non-null     str
 1   Release Year                  10 non-null     int64
 2   Developer                     10 non-null     str
 3   Publisher                     10 non-null     str
 4   Genre                         10 non-null     str
 5   Platform(s)                   10 non-null     str
 6   metacritic_score              10 non-null     int64
 7   estimated_sales_in_millions   10 non-null     float64
 8   ESRB Rating                   10 non-null     str
 9   Avg. Playtime (Hours)         10 non-null     int64
dtypes: float64(1), int64(3), str(6)
memory usage: 932.0 bytes
None

First few rows:
                   Game Title  Release Year         Developer  \
0        Red Dead Redemption 2          2018  Rockstar Studios
1           Grand Theft Auto V          2013    Rockstar North
2   Resident Evil 7: Biohazard          2017            Capcom
...
4                      7

Column names:
['Game Title', 'Release Year', 'Developer', 'Publisher', 'Genre', 'Platform(s)', 'metacritic_score', 'estimated_sales_in_millions', 'ESRB Rating', 'Avg. Playtime (Hours)']
```

What is a Histogram?

A histogram is a chart that plots the distribution of a numeric variable's values as a series of bars. Each bar typically covers a range of numeric values called a bin or class; a bar's height indicates the frequency of data points with a value within the corresponding bin.

```
fig = plt.figure(figsize=(20, 16))

ax1 = plt.subplot(2, 2, 1)
ax1.hist(df['metacritic_score'], bins=8, color='skyblue', edgecolor='black', alpha=0.7)
ax1.set_xlabel('Metacritic Score')
ax1.set_ylabel('Frequency')
ax1.set_title('Distribution of Metacritic Scores', fontsize=14, fontweight='bold')
ax1.grid(True, alpha=0.3)
```

Ouput:

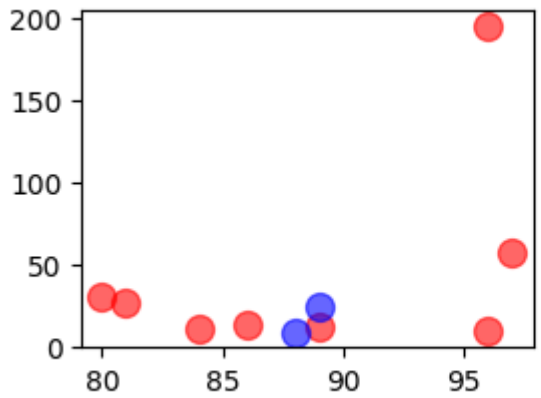


What is a Scatter Plot?

Scatter plots are the graphs that present the relationship between two variables in a data-set. It represents data points on a two-dimensional plane or on a Cartesian system. The independent variable or attribute is plotted on the X-axis, while the dependent variable is plotted on the Y-axis.

```
mean_score = df['metacritic_score'].mean()
median_score = df['metacritic_score'].median()
ax1.axvline(mean_score, color='red', linestyle='dashed', linewidth=2, label=f'Mean:
{mean_score:.1f}')
ax1.axvline(median_score, color='green', linestyle='dashed', linewidth=2, label=f'Median:
{median_score:.1f}')
ax1.legend()

ax2 = plt.subplot(2, 2, 2)

colors = {'M': 'red', 'T': 'blue'}
for rating in df['ESRB Rating'].unique():
    mask = df['ESRB Rating'] == rating
    ax2.scatter(df[mask]['metacritic_score'],
                df[mask]['estimated_sales_in_millions'],
                c=colors[rating],
                label=f'ESRB {rating}',
                s=100,
                alpha=0.6)
```

Ouput:



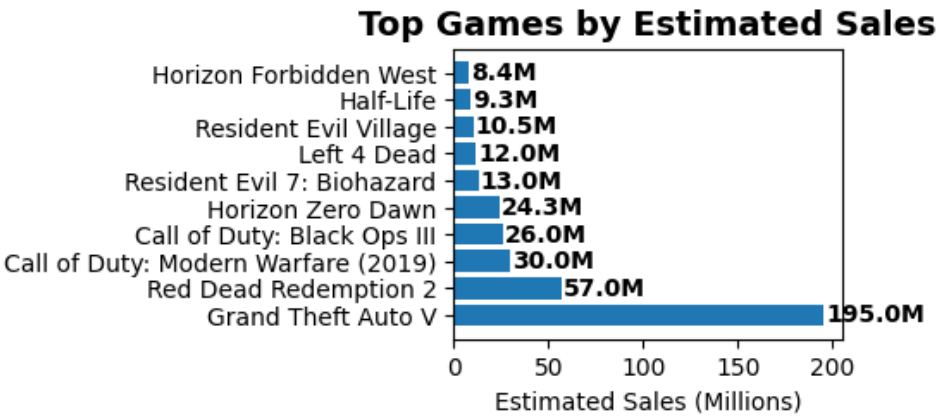
What is a Bar Graph?

A bar graph (or bar chart) is a visual tool that uses rectangular bars to represent and compare numerical data across different categories.

```
ax2.set_xlabel('Metacritic Score')
ax2.set_ylabel('Estimated Sales (Millions)')
ax2.set_title('Metacritic Score vs Estimated Sales', fontsize=14, fontweight='bold')
ax2.legend()
ax2.grid(True, alpha=0.3)
```

```
z = np.polyfit(df['metacritic_score'], df['estimated_sales_in_millions'], 1)
p = np.poly1d(z)
ax2.plot(df['metacritic_score'].sort_values(),
         p(df['metacritic_score'].sort_values()),
         "r--", alpha=0.8, label='Trend Line')

ax3 = plt.subplot(2, 2, 3)
top_games = df.nlargest(10, 'estimated_sales_in_millions')

bars = ax3.barh(range(len(top_games)), top_games['estimated_sales_in_millions'].values)
ax3.set_yticks(range(len(top_games)))
ax3.set_yticklabels(top_games['Game Title'].values, fontsize=10)
ax3.set_xlabel('Estimated Sales (Millions)')
ax3.set_title('Top Games by Estimated Sales', fontsize=14, fontweight='bold')

for i, (bar, sales) in enumerate(zip(bars,
top_games['estimated_sales_in_millions'].values)):
    ax3.text(sales + 1, bar.get_y() + bar.get_height()/2, f'{sales:.1f}M',
             va='center', fontweight='bold')
```

Ouput:



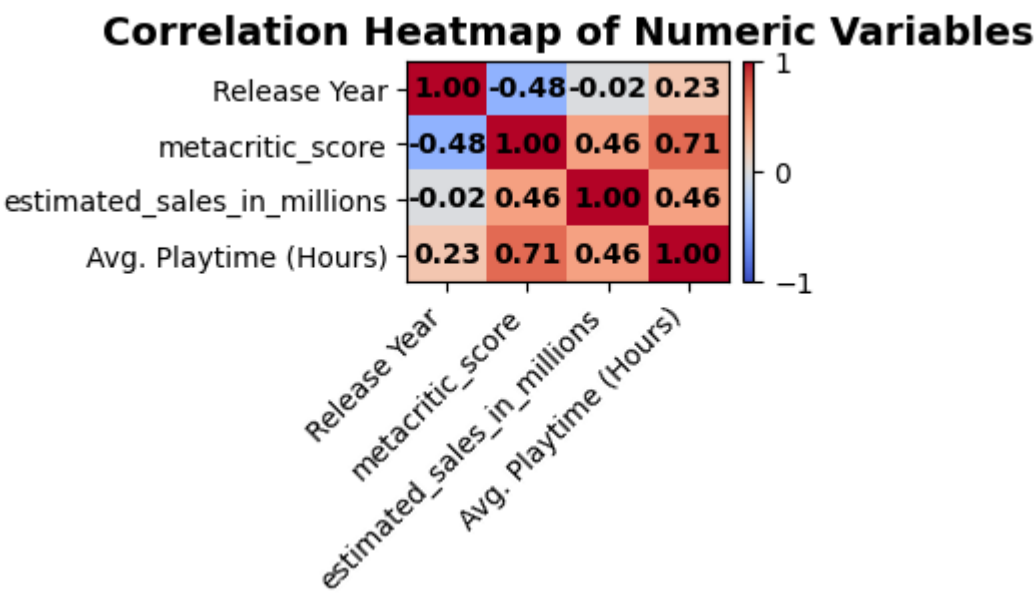
What is a Heatmap?

A heat map is a data visualization technique that represents the magnitude of data points through colors, typically arranged in a 2D grid or map format. Using a spectrum—often warm colors (red/orange) for high intensity and cool colors (blue/purple) for low intensity

```
colors = plt.cm.viridis(np.linspace(0.2, 0.9, len(top_games)))
for bar, color in zip(bars, colors):
    bar.set_color(color)

ax4 = plt.subplot(2, 2, 4)
```

```
numeric_cols = df.select_dtypes(include=[np.number]).columns
correlation_matrix = df[numeric_cols].corr()

im = ax4.imshow(correlation_matrix, cmap='coolwarm', aspect='auto', vmin=-1, vmax=1)

for i in range(len(correlation_matrix.columns)):
    for j in range(len(correlation_matrix.columns)):
        text = ax4.text(j, i, f'{correlation_matrix.iloc[i, j]:.2f}',
                        ha="center", va="center", color="black", fontweight='bold')

ax4.set_xticks(range(len(correlation_matrix.columns)))
ax4.set_yticks(range(len(correlation_matrix.columns)))
ax4.set_xticklabels(correlation_matrix.columns, rotation=45, ha='right')
ax4.set_yticklabels(correlation_matrix.columns)
ax4.set_title('Correlation Heatmap of Numeric Variables', fontsize=14, fontweight='bold')

plt.colorbar(im, ax=ax4, fraction=0.046, pad=0.04)

plt.suptitle('Game Industry Data Analysis Dashboard', fontsize=16, fontweight='bold',
y=1.02)
plt.tight_layout()
plt.show()
```

Ouput:



Conclusion

Successfully demonstrated four key data visualization techniques - histogram, scatter plot, bar chart, and heatmap - enabling effective graphical exploratory data analysis and pattern identification in the dataset.